Ex.No: 01 BASIC NETWORKING COMMANDS
DATE: 23/7/26

AIM:

Study of various network commands used in linux and windows.

1) arp -a :-

Display the address resolution protocol (ARP) table, which contains the IP addresses and their corresponding MAC (physical) addresses stored in the system.

Output:

Interface: 192.168.56.1 ---0xd

Internet Address	Physical address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.2	01-00-5e-00-00-02	Static
224.0.0.22	01-00-5e-00-00-06	Static

2) Hostname :-

Displays the hostname (computer name) of the local machine.

Output

Guruprasad.

3) ipconfig /all : This command displays detailed configuration information about your TCP/IP connection including router, Gateway, DNS, DHCP, and type of Ether adaptor in your system.

Output:

Windows ip configuration

Ethernet adapter vEthernet (WSL-(Hyper-V firewall)):

Connection-specific DNS suffix . :

Link-local IPv6 Address . . : fe80::b44a:e921:20a4

IPv4 Address . . . . . : 172.27.224.1

Subnet Mask . . . . . : 255.255.240.0

4) nbtstat -a : This command helps solve problems with NetBIOS name resolution

Output:

Displays protocol statistics and current TCP/IP connections using NBT (NetBIOS over TCP/IP).

NBTSTAT [[-a RemoteName] [-A IP address] [-c] [-n]
[-r] [-R] [-RR] [-s] [-S] [interval]

-a (adapter status) List the remote machine's name table give its name

-A (adapter status) List the remote machine's name table gives its IP address

5) netstat : (network statistics) netstat displays a variety of statistics about a computers active TCP/IP connections

Output:

Active Connections

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:49680	Guruprasad:49681	ESTABLISHED
TCP	127.0.0.1:49681	Guruprasad:49680	ESTABLISHED
TCP	127.0.0.1:49682	Guruprasad:49683	ESTABLISHED

6) nslookup : Queries the DNS server to obtain the IP address of a domain Name.

Output:

Server: Unknown

Address: 172.16.8.1

Non-authoritative answer:

Name: www.google.com

Addresses: 2001:4860:4829:7700::

2001:4860:482a:7700::

7. Pathping: Combines the features of Ping and Tracert to trace the path to a destination and calculate packet loss at each hop.

Output:

Tracing route to www.google.com [142.251.154.119] over a maximum of 30 hops:

0 Guruprasad [172.16.11.67]
1 172.16.8.1
2 137.97.163.129
3 * * *

8) Ping: Checks whether another computer or website is reachable over the network using ICMP Echo Request and Echo Reply messages

Output:

Pinging www.google.com [142.251.154.119] with 32 bytes of data:
Reply from 142.251.154.119: bytes=32 time = 25 ms TTL = 117
Request timed out.
Reply from 142.251.154.119: bytes=32 time = 24 ms TTL = 117

9) route :

Displays or modifies the IP routing table used by the OS.

Output:

Manipulates network routing tables.

Route [-f] [-P] [-4|-6] Command [destination] [MASK netmask] [gateway] [METRIC metric] [IF interface]

-f clears the routing tables of all gateway entries.

Linux Networking commands

1) ip : Displays and manages IP addresses, routing, and network interfaces.

ip address show

Output:-

1: lo : <LOOPBACK, UP, LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000

link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00

inet 127.0.0.1/8 scope host lo

2) ipconfig :-

Displays network interface details such as IP address, and network statistics

Output:

eth0 : flags = 4163 <UP, BROADCAST, RUNNING, MULTICAST > mtu 1500.

inet 172.17.197.94 netmask 255.255.240.0 broadcast 172.17.205.255

inet 6 fe80 :: 215 : 5dff : fe82 : dc86

prefixlen 64 scopeid 0x20<link>

3) mtr google.com :

Combines ping and Traceroute to display the network path and performance statistics.

Output :

localhost.localdomain (10.0.2.15) → google.com

Host	Loss %	Snt	Last
1. - gateway	0.0%	244	0.6
2. tzdelb-66	0.0%	171	46.3
3. (waiting for reply)			

4) tcpdump -i eth0:

Captures and displays network packets passing through a network interface for analysis.

Output:

tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes

10:15:21.123456 IP 192.168.1.10.52134 > 8.8.8.8.53: UDP, length 32

5) ping google.com

Sends ICMP Echo request packets to verify network connectivity and measure response time

Output:

PING google.com (142.251.43.206) 56 (84) bytes of data.
64 bytes from lcmaaa-bb-in-f14.1e100.net (142.251.43.206): icmp_seq=1 ttl=117 time=5.76 ms

RESULT:

Thus, the various windows and linux networking commands were executed successfully.